

David Xia

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RESEARCH INTERESTS

Theoretical ML, reinforcement learning, online learning, MDP function approximation, sample complexity

EDUCATION

• University of Illinois Urbana-Champaign

Aug 2023 - May 2027

Bachelor of Science in Computer Science

Champaign, Illinois

Bachelor of Science in Liberal Arts & Sciences Major in Mathematics, Data Optimization Concentration

Bachelor of Science in Liberal Arts & Sciences Major in Statistics

◦ GPA: 3.98/4.00

PAPERS

SUB=SUBMITTED, ACC=ACCEPTED

2025 [SUB] RSK linear operators and the Vershik-Kerov-Logan-Shepp curve [arXiv]

Duy Phan, David Xia

Submitted to *Electronic Journal of Combinatorics*

2024 [ACC] EveGuard: Defeating Vibration-based Side-Channel Eavesdropping with Audio Adversarial Perturbations [arXiv]

Jung-Woo Chang, Ke Sun, David Xia, Xinyu Zhang, Farinaz Koushanfar

IEEE Symposium on Security and Privacy, 2025

WORKSHOP PAPERS & MANUSCRIPTS

PRE= IN PREPARATION, SUB=SUBMITTED, ACC=ACCEPTED

2026 [PRE] Frozen Policy Iteration for MDPs with Stochastic Transitions under Linear Q^π Realizability [PDF]

David Xia, Ruizhong Qiu, Hanghang Tong

Manuscript in Preparation

2026 [SUB] Predictability in Major League Sports: Betting Odds versus Mathematical Models [PDF]

David Xia, AJ Hildebrand

Submitted to the *Mathematics and Sports (MAS) Journal*

2026 [ACC] Revitalizing Local Democracy: A Human-Centered Audit of LLMs in City Council Journalism [PDF]

David Xia, Chris Maury

Human-centered Evaluation and Auditing of Language Models (HEAL) Workshop at CHI 2026

RESEARCH EXPERIENCE

• iDEA-iSAIL Lab, Research Intern

Feb 2026 - present

◦ Advised by Professor [Hanghang Tong](#).

◦ Conducted research in theoretical reinforcement learning, studying sample and computation efficient algorithms for online learning in finite-horizon Markov Decision Processes under function approximation and stochastic dynamics.

◦ Used concentrability to develop the first sample and computation efficient algorithm for online reinforcement learning in stochastic-transition MDPs under the linear Q^π realizability assumption, resolving an open problem left by prior work restricted to deterministic dynamics.

◦ Established a PAC sample complexity guarantee of $\tilde{O}(d^2 H^7 C^* / \epsilon^3)$ by combining techniques from both offline and online reinforcement learning theory, dynamic programming, concentration analysis, and linear function approximation.

◦ [Manuscript in preparation.](#)

- **Illinois Combinatorics Lab for Undergraduate Experience (ICLUE)**, Research Intern Aug 2024 - present
 - Advised by Professor [Alexander Yong](#).
 - Proved a new asymptotic result that the probability that the Schensted insertion algorithm for a uniformly random permutation in S_n exhibits a special "bumping" interaction converges to 1 as n approaches infinity.
 - Derived the result using rigorous probability bounds on the RSK linear operator by leveraging the Vershik-Kerov-Logan-Shepp limit shape curve as the central analytic tool.
 - Developed proofs requiring techniques spanning real analysis (uniform convergence, limit arguments), probability theory (concentration inequalities, convergence in probability), and asymptotic analysis (bounding combinatorial quantities in the large- n regime).
 - Paper in submission to the [Electronic Journal of Combinatorics](#).
- **University of California San Diego**, Research Intern Jun 2023 - Dec 2024
 - Advised by Professor [Xinyu Zhang](#).
 - Developed EveGuard, a software-based defense framework to protect voice privacy from vibrometry-based side channels using adversarial audio by using a perturbation generator model (PGM) to effectively suppress sensor-based eavesdropping while preserving high audio quality.
 - Implemented Eve-GAN, a novel domain translation task for inferring eavesdropped signals, enabling end-to-end training of PGM, utilizing few-shot learning techniques to reduce data collection overhead.
 - Achieved a protection rate of over 97% against audio classifiers, hindering eavesdropped reconstruction.
 - Paper published in the [IEEE Symposium on Security and Privacy 2025](#).
- **Illinois Mathematics Lab (formerly Illinois Geometry Lab)**, Research Intern Jan 2024 - present
 - Advised by Professor AJ Hildebrand.
 - Researched the data science perspective of sports matches, analyzing the predictability of outcomes by comparing prediction accuracies across polls, the betting market, Elo ratings, and mathematical models like Bradley-Terry.
 - Presented findings at Rose-Hulman Undergraduate Math Conference 2024, UIUC Undergraduate Research Symposium 2024, and Joint Mathematics Meeting 2025.
 - Paper in submission to the [Mathematics and Sports \(MAS\) Journal](#).
- **Carnegie Mellon University Human-Computer Interaction Institute**, Research Intern May 2025 - Feb 2026
 - Advised by Professor [Jeff Bigham](#).
 - Designed and evaluated an automated journalism pipeline using SOTA LLMs to replicate end-to-end editorial workflows, from transcript segmentation to headline generation and topic prioritization.
 - Orchestrated large-scale crowd-sourced study and demonstrated that LLM-based headline quality and topic prioritization can meet and exceed professional standards.
 - Paper in CHI 2026 [Human-centered Evaluation and Auditing of Language Models \(HEAL\)](#) workshop.

RELEVANT COURSEWORK

CS: Machine Learning, NLP, Deep Learning, Algorithms, Numerical Methods, Numerical Analysis

Math: Honors Real Analysis, Graph Theory, Combinatorics, Abstract Linear Algebra, Abstract Algebra I–II, Algebraic Combinatorics, Linear Programming

Stats: Stochastic Processes, Applied Bayesian Analysis, Survival Analysis

GRANTS AND AWARDS

2025 CMU HCII REU: NSF 2349558

2025 ICLUE RTG: NSF 1937241